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Degree: B.Tech

Roll no.:.....
Semester: 4

End Semester Examination, April 2024

Course Code: CMMTC13

Course title: Applied Linear Algebra

Time: 3 hours

Maximum Marks. 50

Note: Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Attempt any two parts from each Question.	Marks	CO
1a	If $S = \{v_1, v_2, \dots, v_n\}$ is a basis of $V(F)$, then show that every element of V can be uniquely expressed as a linear combination of $v_1, v_2, \dots, v_n$.	5	CO1
1b	Let $R[x]$ be a vector space of polynomials over R . Let $W = \{p(x) \in R[x] \mid p(0) = 0, p(1) = 0\}$ and $S_1 = \{x^2 - x, x^3 - x^2, x^4 - x^3, \dots\}$, $S_2 = \{x^3 - x^2, x^4 - x^3, \dots\}$. Then prove that W is spanned by S_1 but not by S_2 .	5	CO1
1c	Let V be a vector space of all continuous functions on R under usual addition and scalar multiplication over the field R . Let $S_1 = \{\sin(x), \sin(x-1), \sin(x-2)\}$ and $S_2 = \{e^x, e^{x-1}, e^{x-2}\}$. Check if S_1 and S_2 are linearly dependent or linear independent. Justify your answer.	5	CO1
2a	Let $T : R^3 \rightarrow R^3$ be a linear transformation defined by $T(x, y, z) = (x + y, y + z, z + x)$. Is T invertible? If yes, then find $T^{-1}(x, y, z)$.	5	CO2
2b	consider the linear transformation $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + y + 2z, -x + 2y + z, y + 3z)$. Find the matrix of $[T]_{\beta, \beta'}$ and $[T]_{\beta', \beta}$ where, $\beta = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and $\beta' = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$.	5	CO2
2c	Show that $f : R^4 \rightarrow R^4$ given by $f(x, y, z, t) = (2x, 3y, 0, 0)$ is a linear transformation. Find its kernel, range space, rank, and nullity.	5	CO2
3a	Is the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -6 \\ 2 & -2 & 3 \end{bmatrix}$ diagonalizable? If yes, find a matrix P such that $P^{-1}AP$ is a diagonal matrix.	5	CO3
3b	(a) let V is a finite dimensional vector space. and C is an eigenvalue of a linear operator $T : V \rightarrow V$, and $f(x)$ any polynomial over F then show that $f(C)$ is an eigenvalue of $f(T)$, i. e., $f(T)v = f(C)v$, where $T(v) = cv; v \neq 0$. (b) Show that idempotent matrix of order $n \times n$ is always diagonalizable.	5	CO3
3c	Let A be 4×4 real matrix $A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ Find the characteristic polynomial, minimal polynomial for A and its eigen values. Also write the Jordan canonical form of A .	5	CO3

4a	Given $x_1 = (1, 1, 1), x_2 = (1, 2, 3), x_3 = (1, 3, 4) \in \mathbb{R}^3$. Find $\alpha \in \mathbb{R}^3$ such that $\langle \alpha, x_1 \rangle = 7, \langle \alpha, x_2 \rangle = 16, \langle \alpha, x_3 \rangle = 22$.	5	CO4
4b	Let V be a vector space of all real polynomials of degree at most 2 and β is standard basis of V . Find orthonormal basis for V , using Gram Schmidt Process, if the inner product is defined by $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$.	5	CO4
4c	(i) If V is vector space and $\{v_1, v_2, \dots, v_n\}$ is an orthonormal set in V, for each $w \in V$ show that $u = w - \sum_{i=1}^n \langle w, v_i \rangle v_i$ is orthogonal to each of $v_1, v_2, \dots, v_n$. (ii) Show that in a complex inner product space V, if x is orthogonal to y, then $\ x+y\ ^2 = \ x\ ^2 + \ y\ ^2$. Is the converse true? If not, provide a counter example.	5	CO4
5a	Consider the quadratic form $q(x)$ on $\mathbb{R}^3$ given by $q(x) = x_1^2 - x_2^2 + 3x_1x_2 - 6x_2x_3, \forall x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Then write the matrix of quadratic form and classify the definite form of the given quadratic form.	5	CO5
5b	Let $V(\mathbb{R})$ is a vector space of 3×3 real symmetric matrices. Define a function $f : V \times V \rightarrow \mathbb{R}$ such that $f(A, B) = \text{Trace}(AB)$. Prove that given function is bilinear form but not an inner product.	5	CO5
5c	Consider $f : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(u, v) = x_1y_1 - \alpha(x_1y_2 + x_2y_1) + \beta x_2y_2$, where $u = (x_1, x_2)^T, v = (y_1, y_2)^T \in \mathbb{R}^2$. Find the necessary and sufficient conditions on α, β such that quadratic form $f(u, u)$ is positive definite.	5	CO5